THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — MATHS · EASY LEVEL CHAPTER TEST SET 2

Chapter 7: **Probability** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A bag has 4 red, 6 blue and 5 green balls. The probability of drawing a green ball is: 1

(a) $\frac{1}{3}$ **(b)** $\frac{1}{4}$ **(c)** $\frac{2}{3}$ **(d)** $\frac{1}{6}$

2. A ticket is drawn from tickets numbered 1 to 20. The probability that the number is a multiple of 5 is: 1

(a) $\frac{4}{5}$ **(b)** $\frac{1}{5}$ **(c)** $\frac{1}{6}$ **(d)** $\frac{2}{5}$

3. The probability of a sure (certain) event is: 1

(a) $\frac{1}{6}$ **(b)** 1 **(c)** 2 **(d)** $\frac{1}{2}$

4. The number of outcomes when a coin is tossed once is: 1

(a) 4 **(b)** 1 **(c)** 2 **(d)** 3

5. If $P(E) = \frac{3}{8}$, then $P(\text{not } E)$ is: 1

(a) $\frac{3}{4}$ **(b)** $\frac{5}{9}$ **(c)** $\frac{5}{8}$ **(d)** $\frac{3}{8}$

6. A coin is tossed 50 times and a head appears 27 times. The experimental probability of a head is: 1

(a) $\frac{14}{25}$ **(b)** $\frac{23}{50}$ **(c)** $\frac{27}{50}$ **(d)** $\frac{9}{17}$

7. A fair die is rolled once. The probability of getting a multiple of 3 is: 1

(a) $\frac{1}{4}$ **(b)** $\frac{2}{3}$ **(c)** $\frac{1}{6}$ **(d)** $\frac{1}{3}$

8. A bag contains 3 red and 5 blue balls. A ball is drawn at random. The probability that it is red is: 1

(a) $\frac{3}{8}$ **(b)** $\frac{1}{3}$ **(c)** $\frac{5}{8}$ **(d)** $\frac{1}{2}$

9. A bag has 5 red, 7 blue and 8 green balls. The probability of drawing a green ball is: 1

(a) $\frac{1}{3}$ **(b)** $\frac{1}{5}$ **(c)** $\frac{3}{5}$ **(d)** $\frac{2}{5}$

10. A ticket is drawn from tickets numbered 1 to 12. The probability that the number is a multiple of 3 is: 1

(a) $\frac{1}{6}$ **(b)** $\frac{1}{3}$ **(c)** $\frac{1}{4}$ **(d)** $\frac{2}{3}$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A bag contains 6 red and 9 white balls. Find the probability of drawing a white ball. 2

12. A coin is tossed 150 times and shows heads 72 times. Find the experimental probability of heads and of tails. 2

13. A fair die is rolled. Find the probability of getting a number at least 5. 2

14. A bag contains 7 red and 8 white balls. Find the probability of drawing a red ball. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A bag contains 8 red and 12 blue balls. A ball is drawn at random. Find $P(\text{red})$ and $P(\text{not red})$. 3

16. A die was thrown 90 times and the number 1 appeared 17 times. Find the experimental probability of getting 1, and of not getting 1. 3

17. A ticket is drawn at random from tickets numbered 1 to 40. Find the probability that it shows a multiple of 8. 3

OR

A bag contains 9 red and 6 blue balls. A ball is drawn at random. Find $P(\text{red})$ and $P(\text{not red})$. 3

18. A die was thrown 150 times and the number 3 appeared 28 times. Find the experimental probability of getting 3, and of not getting 3. 3

OR

A ticket is drawn at random from tickets numbered 1 to 36. Find the probability that it shows a multiple of 6. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A bag contains 3 red, 7 blue and 5 green balls. Find $P(\text{red})$, $P(\text{blue})$, $P(\text{green})$ and show that they add up to 1. 5

OR

A fair die is rolled once. Find 5
(i) $P(\text{a multiple of } 3)$,
(ii) $P(\text{a number at most } 2)$ and
(iii) $P(\text{a multiple of } 3 \text{ or a number at most } 2)$ using the favourable outcomes.

20. A bag contains 6 red, 8 blue and 4 green balls. Find $P(\text{red})$, $P(\text{blue})$, $P(\text{green})$ and show that they add up to 1. 5

OR

A fair die is rolled once. Find 5
(i) $P(\text{a prime number})$,
(ii) $P(\text{a perfect square})$ and
(iii) $P(\text{a prime number or a perfect square})$ using the favourable outcomes.

END OF QUESTION PAPER — **Total: 40 Marks**

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